Ex no : 1

Date :

Basic Network Commands

Aim:

To study of various Network commands used in Linux and Windows and to use them for viewing networking configuration, testing connectivity, resolving hostnames, displaying routing information and troubleshooting.
Networking related problem.

Commands :-

1. arp -a : ARP is short form of address resolution protocol. It will show the IP address of your computer along with the IP address and MAC address of your router.

2. hostname : This is the simplest of all TCP/IP commands. It simply displays the name of your computer.

3. ipconfig/all : This command display detailed configuration information about your TCP/IP Connection including Router, Gateway, DNS, DHCP, and type of Ethernet adapter in your system.

4. nbstat -a : This command helps solve problems with NetBIOS name resolution. (Nbt stands for NetBIOS over TCP/IP)

5. netstat : (network statistics) netstat displays a variety of statistics about a computer active TCP/IP connections. It is command line tool for monitoring network connections both incoming and outgoing as well as viewing routing tables, interface statistics etc.

e.g :- netstat -r

6. nslookup (name server lookup) is a tool used to perform DNS lookups in LINUX. It is used to display DNS details, such as the IP address of a particular computer, the MX records for a domain or the NS servers of a domain. nslookup can operate in two modes : interactive and non-interactive.

e.g : nslookup www.google.com

7. pathping : pathping is unique to windows and is basically a combination of the ping and tracert commands. pathping traces

test of each router along the way, gathering statistics on the rate of data loss along each hop.

8. Ping: (Packet Internet Groper) command is the best way to test connectivity between two nodes. Ping use ICMP (Internet Control Message Protocol) to communication to other devices.

1. #ping hostname (Ping localhost)
2. #ping ip address (ping 4.2.2.2)
3. #ping fully qualified domain name (Ping www.facebook.com)

9. Route: route command is used to show/manipulate the IP routing table. It is primarily used to setup static routes to specific hosts or networks via an interface.

LINUX NETWORKING COMMAND

1. Ip: The ip command is one of the basic commands every administrator will need in daily work, from setting up new systems and assigning IPs to troubleshooting existing systems. The ip command can show address information, manipulate routing, plus display network variables devices, interfaces, and tunnels.

IP <OPTIONS> <OBJECT> <COMMAND>

2. Ifconfig: The ifconfig command used is a staple in many sysadmin's tool belt for configuring and troubleshooting networks. It has since been replaced by the ip command discussed above

3. mtr: MTR (Matt's traceroute) is a program with a command-line interface that serves as a network diagnostic and troubleshooting tool. This command combines the functionality of the Ping and traceroute commands. Just like a trace route, the mtr command will show the route from a computer to a specified host.

Syntax:
mtr <options> hostname/IP

4) tcpdump: The tcpdump command is designed for capturing and displaying packets.

You can install tcpdump with the command
[root@server ~]# dnf install -y tcpdump

5) ping: Ping is a tool that verifies IP-level connectivity to another TCP/IP computer by sending Internet Control Message Protocol (ICMP) Echo Request messages. The receipt of corresponding Echo Reply message is displayed, along with round-trip times. Ping is the primary TCP/IP command used to troubleshoot connectivity, reachability and name resolution.

Commands

1. IP address show
2. mtr google.com
3. mtr gauge.com
4. tcpdump -D
5. ping gauge.

i) arp -a

```
C:\Users\DELL>arp -a

Interface: 192.168.56.1 --- 0x10
  Internet Address      Physical Address      Type
  192.168.56.255        ff-ff-ff-ff-ff-ff     static
  224.0.0.2             01-00-5e-00-00-02     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  224.0.0.253           01-00-5e-00-00-fd     static
  239.0.0.8             01-00-5e-00-00-08     static
  239.192.152.143       01-00-5e-40-98-8f     static
  239.255.102.18        01-00-5e-7f-66-12     static
  239.255.255.250       01-00-5e-7f-ff-fa     static

Interface: 172.16.75.31 --- 0x12
  Internet Address      Physical Address      Type
  172.16.72.1           7c-5a-1c-cf-be-41     dynamic
  172.16.73.38          f8-3d-c6-d5-74-ac     dynamic
  172.16.73.125         dc-45-46-bf-42-26     dynamic
  172.16.73.141         38-ca-84-3d-5e-a0     dynamic
  172.16.73.197         7c-57-58-37-2d-d8     dynamic
  172.16.75.1           4c-82-a9-77-ff-db     dynamic
  172.16.75.4           4c-82-a9-78-00-9d     dynamic
  172.16.75.5           4c-82-a9-77-ff-61     dynamic
  172.16.75.8           4c-82-a9-79-19-99     dynamic
  172.16.75.10          4c-82-a9-77-fe-fd     dynamic
  172.16.75.11          4c-82-a9-77-fe-e7     dynamic
  172.16.75.16          4c-82-a9-77-ff-f5     dynamic
  172.16.75.17          4c-82-a9-77-ff-19     dynamic
```

ii) hostname

```
C:\Users\DELL>hostname
HDC0124091

C:\Users\DELL>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet 2:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::b0f:7aab:72d:9bb3%16
   IPv4 Address. . . . . . . . . . . : 192.168.56.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Wireless LAN adapter Local Area Connection* 9:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 10:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter WiFi:

   Connection-specific DNS Suffix  . :
   IPv4 Address. . . . . . . . . . . : 172.16.75.31
   Subnet Mask . . . . . . . . . . . : 255.255.248.0
   Default Gateway . . . . . . . . . : 172.16.72.1

C:\Users\DELL>ipconfigall
'ipconfigall' is not recognized as an internal or external command,
operable program or batch file.
```

iii) nbstat -a

```
C:\Users\DELL>nbtstat -a

Displays protocol statistics and current TCP/IP connections using NBT
(NetBIOS over TCP/IP).

NBTSTAT [ [-a RemoteName] [-A IP address] [-c] [-n]
        [-r] [-R] [-RR] [-s] [-S] [interval] ]

  -a   (adapter status) Lists the remote machine's name table given its name
  -A   (Adapter status) Lists the remote machine's name table given its
                        IP address.
  -c   (cache)          Lists NBT's cache of remote [machine] names and their IP addresses
  -n   (names)          Lists local NetBIOS names.
  -r   (resolved)       Lists names resolved by broadcast and via WINS
  -R   (Reload)         Purges and reloads the remote cache name table
  -S   (Sessions)       Lists sessions table with the destination IP addresses
  -s   (sessions)       Lists sessions table converting destination IP
                        addresses to computer NETBIOS names.
  -RR  (ReleaseRefresh) Sends Name Release packets to WINS and then, starts Refresh

  RemoteName   Remote host machine name.
  IP address   Dotted decimal representation of the IP address.
  interval     Redisplays selected statistics, pausing interval seconds
               between each display. Press Ctrl+C to stop redisplaying
               statistics.
```

iii) Ipconfig all

```
C:\Users\DELL>ipconfig all

Error: unrecognized or incomplete command line.

USAGE:
    ipconfig [/allcompartments] [/? | /all |
                                 /renew [adapter] | /release [adapter] |
                                 /renew6 [adapter] | /release6 [adapter] |
                                 /flushdns | /displaydns | /registerdns |
                                 /showclassid adapter |
                                 /setclassid adapter [classid] |
                                 /showclassid6 adapter |
                                 /setclassid6 adapter [classid] ]

where
    adapter             Connection name
                        (wildcard characters * and ? allowed, see examples)

    Options:
       /?               Display this help message
       /all             Display full configuration information.
       /release         Release the IPv4 address for the specified adapter.
       /release6        Release the IPv6 address for the specified adapter.
       /renew           Renew the IPv4 address for the specified adapter.
       /renew6          Renew the IPv6 address for the specified adapter.
       /flushdns        Purges the DNS Resolver cache.
       /registerdns     Refreshes all DHCP leases and re-registers DNS names
       /displaydns      Display the contents of the DNS Resolver Cache.
       /showclassid     Displays all the dhcp class IDs allowed for adapter.
       /setclassid      Modifies the dhcp class id.
       /showclassid6    Displays all the IPv6 DHCP class IDs allowed for adapter.
       /setclassid6     Modifies the IPv6 DHCP class id.

The default is to display only the IP address, subnet mask and
default gateway for each adapter bound to TCP/IP.

For Release and Renew, if no adapter name is specified, then the IP address
leases for all adapters bound to TCP/IP will be released or renewed.

For Setclassid and Setclassid6, if no ClassId is specified, then the ClassId is removed.

Examples:
    > ipconfig                  ... Show information
    > ipconfig /all             ... Show detailed information
    > ipconfig /renew           ... renew all adapters
    > ipconfig /renew EL*       ... renew any connection that has its
                                    name starting with EL
    > ipconfig /release *Con*   ... release all matching connections,
                                    eg. "Wired Ethernet Connection 1" or
                                        "Wired Ethernet Connection 2"
```

iv) netstat

```
C:\Users\DELL>netstat

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    127.0.0.1:49672        HDC0124091:49673       ESTABLISHED
  TCP    127.0.0.1:49673        HDC0124091:49672       ESTABLISHED
  TCP    127.0.0.1:49674        HDC0124091:49675       ESTABLISHED
  TCP    127.0.0.1:49675        HDC0124091:49674       ESTABLISHED
  TCP    127.0.0.1:49677        HDC0124091:49678       ESTABLISHED
  TCP    127.0.0.1:49678        HDC0124091:49677       ESTABLISHED
  TCP    172.16.75.31:7680      172.16.75.24:57351     TIME_WAIT
  TCP    172.16.75.31:7680      172.16.75.53:52151     ESTABLISHED
  TCP    172.16.75.31:7680      172.16.75.57:56433     TIME_WAIT
  TCP    172.16.75.31:7680      172.16.75.105:52729    ESTABLISHED
  TCP    172.16.75.31:7680      172.16.75.105:55964    TIME_WAIT
  TCP    172.16.75.31:7680      172.16.75.106:54219    TIME_WAIT
  TCP    172.16.75.31:7680      172.16.75.214:59506    ESTABLISHED
  TCP    172.16.75.31:7680      172.16.75.214:59827    TIME_WAIT
  TCP    172.16.75.31:50169     pnmaaa-aq-in-f5:https  ESTABLISHED
  TCP    172.16.75.31:50248     se-in-f188:5228        ESTABLISHED
  TCP    172.16.75.31:52020     se-in-f188:5228        ESTABLISHED
  TCP    172.16.75.31:55096     13.107.6.254:https     ESTABLISHED
  TCP    172.16.75.31:55496     104.18.39.21:https     ESTABLISHED
  TCP    172.16.75.31:55961     52.108.8.254:https     ESTABLISHED
  TCP    172.16.75.31:56229     sin10s07-in-f106:https ESTABLISHED
  TCP    172.16.75.31:56552     4.150.245.254:https    ESTABLISHED
  TCP    172.16.75.31:56975     204.79.197.222:https   ESTABLISHED
  TCP    172.16.75.31:57457     4.213.25.241:https     ESTABLISHED
  TCP    172.16.75.31:58106     172.187.55.27:https    ESTABLISHED
  TCP    172.16.75.31:58108     172.16.77.47:ms-do     SYN_SENT
  TCP    172.16.75.31:61484     172.16.75.36:ms-do     ESTABLISHED
  TCP    172.16.75.31:62773     172.64.155.209:https   ESTABLISHED
  TCP    172.16.75.31:63318     pnmaaa-aq-in-f5:https  TIME_WAIT
  TCP    172.16.75.31:65195     150.171.70.254:https   ESTABLISHED
  TCP    [fe80::b0f:7aab:72d:9bb3%16]:1521  HDC0124091:58170       ESTABLISHED
  TCP    [fe80::b0f:7aab:72d:9bb3%16]:58170  HDC0124091:1521       ESTABLISHED
```

vii) pathping

```
C:\Users\DELL>pathping

Usage: pathping [-g host-list] [-h maximum_hops] [-i address] [-n]
                [-p period] [-q num_queries] [-w timeout]
                [-4] [-6] target_name

Options:
    -g host-list     Loose source route along host-list.
    -h maximum_hops  Maximum number of hops to search for target.
    -i address       Use the specified source address.
    -n               Do not resolve addresses to hostnames.
    -p period        Wait period milliseconds between pings.
    -q num_queries   Number of queries per hop.
    -w timeout       Wait timeout milliseconds for each reply.
    -4               Force using IPv4.
    -6               Force using IPv6.

C:\Users\DELL>
```

iv) netstat -r

```
C:\Users\DELL>netstat -r
===========================================================================
Interface List
  9...20 88 10 86 8c 68 ......Intel(R) Ethernet Connection (17) I219-LM
 16...0a 00 27 00 00 10 ......VirtualBox Host-Only Ethernet Adapter
 14...4e 82 a9 78 00 0b ......Microsoft Wi-Fi Direct Virtual Adapter
  3...42 82 a9 78 80 0b ......Microsoft Wi-Fi Direct Virtual Adapter #2
 18...4c 82 a9 78 00 0b ......Realtek RTL8852BE WiFi 6 802.11ax PCIe Adapter
  1...........................Software Loopback Interface 1
===========================================================================

IPv4 Route Table
===========================================================================
Active Routes:
Network Destination        Netmask          Gateway       Interface  Metric
          0.0.0.0          0.0.0.0      172.16.72.1    172.16.75.31     286
        127.0.0.0        255.0.0.0         On-link         127.0.0.1     331
        127.0.0.1  255.255.255.255         On-link         127.0.0.1     331
  127.255.255.255  255.255.255.255         On-link         127.0.0.1     331
       172.16.72.0    255.255.248.0         On-link    172.16.75.31     286
      172.16.75.31  255.255.255.255         On-link    172.16.75.31     286
    172.16.79.255  255.255.255.255         On-link    172.16.75.31     286
      192.168.56.0    255.255.255.0         On-link    192.168.56.1     281
      192.168.56.1  255.255.255.255         On-link    192.168.56.1     281
    192.168.56.255  255.255.255.255         On-link    192.168.56.1     281
        224.0.0.0        240.0.0.0         On-link         127.0.0.1     331
        224.0.0.0        240.0.0.0         On-link    192.168.56.1     281
        224.0.0.0        240.0.0.0         On-link    172.16.75.31     286
  255.255.255.255  255.255.255.255         On-link         127.0.0.1     331
  255.255.255.255  255.255.255.255         On-link    192.168.56.1     281
  255.255.255.255  255.255.255.255         On-link    172.16.75.31     286
===========================================================================
Persistent Routes:
  Network Address          Netmask  Gateway Address  Metric
          0.0.0.0          0.0.0.0      172.16.72.1  Default
===========================================================================

IPv6 Route Table
===========================================================================
Active Routes:
 If Metric Network Destination      Gateway
  1    331 ::1/128                  On-link
 16    281 fe80::/64                On-link
 16    281 fe80::b0f:7aab:72d:9bb3/128
                                    On-link
  1    331 ff00::/8                 On-link
 16    281 ff00::/8                 On-link
===========================================================================
Persistent Routes:
  None
```

ix) Ping

```
C:\Users\DELL>Route

Manipulates network routing tables.

ROUTE [-f] [-p] [-4|-6] command [destination]
                  [MASK netmask]  [gateway] [METRIC metric]  [IF interface]

  -f           Clears the routing tables of all gateway entries.  If this is
               used in conjunction with one of the commands, the tables are
               cleared prior to running the command.

  -p           When used with the ADD command, makes a route persistent across
               boots of the system. By default, routes are not preserved
               when the system is restarted. Ignored for all other commands,
               which always affect the appropriate persistent routes.

  -4           Force using IPv4.

  -6           Force using IPv6.

  command      One of these:
                 PRINT     Prints  a route
                 ADD       Adds    a route
                 DELETE    Deletes a route
                 CHANGE    Modifies an existing route
  destination  Specifies the host.
  MASK         Specifies that the next parameter is the 'netmask' value.
  netmask      Specifies a subnet mask value for this route entry.
               If not specified, it defaults to 255.255.255.255.
  gateway      Specifies gateway.
  interface    the interface number for the specified route.
  METRIC       specifies the metric, ie. cost for the destination.

All symbolic names used for destination are looked up in the network database
file NETWORKS. The symbolic names for gateway are looked up in the host name
database file HOSTS.

If the command is PRINT or DELETE. Destination or gateway can be a wildcard,
(wildcard is specified as a star '*'), or the gateway argument may be omitted.

If Dest contains a * or ?, it is treated as a shell pattern, and only
matching destination routes are printed. The '*' matches any string,
and '?' matches any one char. Examples: 157.*.1, 157.*, 127.*, *224*.

Pattern match is only allowed in PRINT command.
Diagnostic Notes:
    Invalid MASK generates an error, that is when (DEST & MASK) != DEST.
    Example> route ADD 157.0.0.0 MASK 155.0.0.0 157.55.80.1 IF 1
             The route addition failed: The specified mask parameter is invalid. (Destination & Mask) != Destination.

Examples:

    > route PRINT
    > route PRINT -4
    > route PRINT -6
    > route PRINT 157*            .... Only prints those matching 157*

    > route ADD 157.0.0.0 MASK 255.0.0.0  157.55.80.1 METRIC 3 IF 2
              destination^    ^mask      ^gateway     metric^   ^
                                                       Interface^
      If IF is not given, it tries to find the best interface for a given
      gateway.
    > route ADD 3ffe::/32 3ffe::1

    > route CHANGE 157.0.0.0 MASK 255.0.0.0 157.55.80.5 METRIC 2 IF 2

      CHANGE is used to modify gateway and/or metric only.

    > route DELETE 157.0.0.0
    > route DELETE 3ffe::/32
```

vi) nslookup

```
:\Users\DELL>nslookup www.google.com
erver:  UnKnown
ddress:  172.16.72.1

Non-authoritative answer:
Name:    www.google.com
Addresses:  2001:4860:4826:7700::
          2001:4860:482c:7700::
          2001:4860:4828:7700::
          2001:4860:482a:7700::
          2001:4860:482b:7700::
          2001:4860:4829:7700::
          2001:4860:482d:7700::
          2001:4860:4827:7700::
          142.251.150.119
          142.251.155.119
          142.251.152.119
          142.251.154.119
          142.251.157.119
          142.251.151.119
          142.251.156.119
          142.251.153.119
```

vii) ping

```
C:\Users\DELL>ping

Usage: ping [-t] [-a] [-n count] [-l size] [-f] [-i TTL] [-v TOS]
            [-r count] [-s count] [[-j host-list] | [-k host-list]]
            [-w timeout] [-R] [-S srcaddr] [-c compartment] [-p]
            [-4] [-6] target_name

Options:
    -t             Ping the specified host until stopped.
                   To see statistics and continue - type Control-Break;
                   To stop - type Control-C.
    -a             Resolve addresses to hostnames.
    -n count       Number of echo requests to send.
    -l size        Send buffer size.
    -f             Set Don't Fragment flag in packet (IPv4-only).
    -i TTL         Time To Live.
    -v TOS         Type Of Service (IPv4-only. This setting has been deprecated
                   and has no effect on the type of service field in the IP
                   Header).
    -r count       Record route for count hops (IPv4-only).
    -s count       Timestamp for count hops (IPv4-only).
    -j host-list   Loose source route along host-list (IPv4-only).
    -k host-list   Strict source route along host-list (IPv4-only).
    -w timeout     Timeout in milliseconds to wait for each reply.
    -R             Use routing header to test reverse route also (IPv6-only).
                   Per RFC 5095 the use of this routing header has been
                   deprecated. Some systems may drop echo requests if
                   this header is used.
    -S srcaddr     Source address to use.
    -c compartment Routing compartment identifier.
    -p             Ping a Hyper-V Network Virtualization provider address.
    -4             Force using IPv4.
    -6             Force using IPv6.
```

Linux Commands

i) ip route

```
rtt min/avg/max/mdev = 7.693/9.699/15.866/2.436 ms
student@fedora:~$ ip route
default via 10.0.2.2 dev enp0s3 proto dhcp src 10.0.2.15 metric 100

10.0.2.0/24 dev enp0s3 proto kernel scope link src 10.0.2.15 metric
 100
student@fedora:~$
```

ii) ip addr show

```
student@fedora:~$ ip  addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN
 group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_code
l state UP group default qlen 1000
    link/ether 08:00:27:2d:91:bf brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixr
oute enp0s3
       valid_lft 84400sec preferred_lft 84400sec
    inet6 fd17:625c:f037:2:eebe:9b47:404f:31ac/64 scope global dyna
mic noprefixroute
       valid_lft 86126sec preferred_lft 14126sec
    inet6 fe80::55f0:9771:cb55:7bf3/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
student@fedora:~$
```

Click to switch to "Workspace 4"

iii)

```
                    My traceroute  [v0.95]
fedora (10.0.2.15) -> google.com (142.250.2026-07-16T08:51:40+0530
Keys:  Help   Display mode   Restart statistics   Order of fields
quit                          Packets               Pings
Host                        Loss%   Snt   Last   Avg  Best  Wrst StDev
1. pnmaaa-aq-in-f14.1e   0.0%    16    8.0  10.6   7.2  16.6   3.0
```

iii) tcpdump -ienp0s3

```
0 packets dropped by kernel
root@fedora:/home/student# tcpdump -ienp0s3
dropped privs to tcpdump
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on enp0s3, link-type EN10MB (Ethernet), snapshot length 262144 bytes
14:19:32.526843 IP fedora.56049 > www.time.nplindia.org.ntp: NTPv4, Client, leng
th 48
14:19:32.649966 IP www.time.nplindia.org.ntp > fedora.56049: NTPv4, Server, leng
th 48
^C
2 packets captured
2 packets received by filter
0 packets dropped by kernel
root@fedora:/home/student#
```

iv) ip a

```
0 enp0s3
student@fedora: $ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN
group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_code
l state UP group default qlen 1000
    link/ether 08:00:27:2d:91:bf brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixr
oute enp0s3
       valid_lft 84984sec preferred_lft 84984sec
    inet6 fd17:625c:f037:2:eebe:9b47:404f:31ac/64 scope global dyna
mic noprefixroute
       valid_lft 86199sec preferred_lft 14199sec
    inet6 fe80::55f0:9771:cb55:7bf3/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
student@fedora: $
```

v) tcpdump -ienp0s3 -c 10

```
root@fedora:/home/student# ^C
root@fedora:/home/student# tcpdump -ienp0s3 -c 10
dropped privs to tcpdump
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on enp0s3, link-type EN10MB (Ethernet), snapshot length 262144 bytes
14:28:06.127023 IP6 _gateway > ff02::1: ICMP6, router advertisement, length 56
14:28:06.130404 IP6 fedora > ff02::16: HBH ICMP6, multicast listener report v2, 4 group record(s), length 88
14:28:06.201018 IP fedora.58502 > 172.16.72.1.domain: 58692+ PTR? 1.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0
.0.0.2.0.f.f.ip6.arpa. (90)
14:28:06.280127 IP 172.16.72.1.domain > fedora.58502: 58692 NXDomain* 0/0/0 (90)
14:28:06.425811 IP fedora.42919 > 172.16.72.1.domain: 42807+ PTR? 6.1.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0
.0.0.2.0.f.f.ip6.arpa. (90)
14:28:06.438088 IP 172.16.72.1.domain > fedora.42919: 42807 NXDomain* 0/0/0 (90)
14:28:06.494433 IP6 fedora > ff02::16: HBH ICMP6, multicast listener report v2, 4 group record(s), length 88
14:28:06.675900 IP fedora.40294 > 172.16.72.1.domain: 22200+ PTR? 1.72.16.172.in-addr.arpa. (42)
14:28:06.689865 IP 172.16.72.1.domain > fedora.40294: 22200 NXDomain* 0/0/0 (42)
14:28:06.923580 IP fedora.51816 > 172.16.72.1.llmnr: Flags [S], seq 1231455704, win 64240, options [mss 1460,sackOK,T
S val 3476406632 ecr 0,nop,wscale 7,tfo  cookiereq,nop,nop], length 0
10 packets captured
10 packets received by filter
0 packets dropped by kernel
root@fedora:/home/student# tcpdump -i
```

vi) tcpdump -ienp0s3

```
(No such device exists)
root@fedora:/home/student# tcpdump -ienp0s3
dropped privs to tcpdump
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on enp0s3, link-type EN10MB (Ethernet), snapshot length 262144 bytes
14:15:12.466438 IP fedora.38142 > mumbai-41.gombadi.com.ntp: NTPv4, Client, leng
th 48
14:15:12.528783 IP fedora.47066 > 172.16.72.1.domain: 4417+ [1au] PTR? 33.16.113
.148.in-addr.arpa. (55)
14:15:12.547076 IP mumbai-41.gombadi.com.ntp > fedora.38142: NTPv4, Server, leng
th 48
14:15:12.547077 IP 172.16.72.1.domain > fedora.47066: 4417 1/0/0 PTR mumbai-41.g
ombadi.com. (79)
14:15:12.547394 IP fedora.47066 > 172.16.72.1.domain: 34981+ PTR? 33.16.113.148.
in-addr.arpa. (44)
14:15:12.549387 IP fedora.37912 > mumbai-41.gombadi.com.llmnr: Flags [S], seq 41
97231118, win 64240, options [mss 1460,sackOK,TS val 399930269 ecr 0,nop,wscale
7,tfo  cookiereq,nop,nop], length 0
14:15:12.579746 IP 172.16.72.1.domain > fedora.47066: 34981 1/0/0 PTR mumbai-41.
gombadi.com. (79)
```

Result:

The basic commands for linux & window are successfully applied.